UGV

Progress Log

Materials

ESP32 devkit,chassis,2 side wheels, caster wheel,L293 motor driver IC, breadboard(small), jumper wires, power bank, micro usb to usb A wire. DC motors.

Progress Log

– We are told about this project with a very few days to understand and build the UGV.

– We went through the pdf and understood the aim of this project and also we had the clear picture of what we needed to do.

–We got the required materials to build the project and started working on it immediately as there was very little time. We received a power bank instead of a normal Lipo battery pack because there was a shortage of those batteries.

–We used AI to make changes to the existing code so that we can start experimenting on the vehicle and to make sure that everything works with the restricted power bank supply.

–We have used VS code for flashing the ESP and also to tweak the code at a later point of time.

–We have successfully assembled the entire vehicle and uploaded the code. But there were some wrong connections. We accidentally shorted the 3.3V supply to the 5V supply and fortunately there was no damage to the ESP.

– We faced many difficulties with connecting the ESP to the phone via bluetooth. The ESP was not responding to bluetooth. This was all due to some more improper connections that we fixed later.

– We started testing the vehicle to finetune the code so that it works perfectly.

–We began to realize a problem. The 5V of the power bank was not enough to power the motors. They are reliably turning on under load only at almost 5V. This left us very little room for tuning the motors via PWM.

–We continued to push through even though we knew that it was not going to be perfect. We got a fairly good result with the corrections we made, but still, it was not perfect.

–We tested the vehicle again after a day and this time the vehicle needed correction again. This was primarily due to the motor defects and we could not do much about it except tune it again.

– We cannot be sure that the code provided will perfectly work with the vehicle after a few days but we can’t do anything about it.

https://youtu.be/z5iIIcam-js







